

# Lab 1: Building Your Isolated Ethical Hacking Lab Environment

## Session 1 — Tool & Software Setup

**Course:** Hands-On Ethical Hacking and Network Defense, 4e — Module 1 companion lab **Estimated time:** 90-120 minutes **Based on:** Module 1, “Your Lab This Semester” and “Your Lab, On Screen”

### 1. Learning Objectives

By the end of this session you will be able to:

- Install and configure Oracle VirtualBox as your virtualization platform
- Import and correctly network the three VMs used throughout this course: **Kali Linux** (attacker), **Metasploitable 2** (target), and **OWASP BWA** (target)
- Build an isolated **host-only network** so attack traffic can never reach the campus network or the Internet
- Take and understand VM snapshots
- Verify your environment matches the reference architecture before running a single tool

### 2. ⚠ Read Before You Begin

Metasploitable 2 and OWASP BWA are **deliberately vulnerable** and ship with unauthenticated remote code execution. Exposing either one to a real network — campus Wi-Fi, home network, or the Internet — is a security incident, not a classroom mistake. Every step below exists to prevent that.

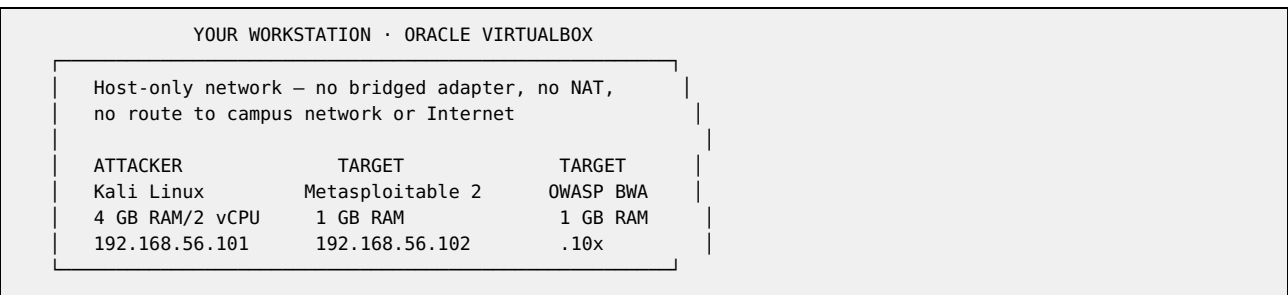
This lab only touches systems inside your own host-only network, which you own and control, so no written authorization is required for this exercise. Anything outside this box requires the written authorization Module 1 describes — never point any tool built in this lab at a network you don’t have permission to test.

### 3. Software & Files You’ll Need

| Item                                               | Purpose                   | Notes                                                                                  |
|----------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------|
| Oracle VirtualBox (latest stable) + Extension Pack | Hypervisor                | <a href="https://www.virtualbox.org/wiki/Downloads">virtualbox.org/wiki/Downloads</a>  |
| Kali Linux VM image (VirtualBox build)             | Attacker platform         | <a href="https://kali.org/get-kali">kali.org/get-kali</a> — official VirtualBox images |
| Metasploitable 2 image                             | Vulnerable Linux target   | as provided/linked by your instructor                                                  |
| OWASP Broken Web Apps (BWA) VM                     | Vulnerable web-app target | as provided/linked by your instructor                                                  |
| Host machine resources                             | —                         | 20 GB free disk, 8 GB RAM recommended                                                  |

Confirm with your instructor which exact download links or internal mirrors are approved before pulling anything from the open internet.

### 4. Reference Architecture



The three VMs can reach each other and nothing else.

### 5. Procedure

## Part A — Install VirtualBox

1. Download and install VirtualBox for your host OS.
2. Install the matching VirtualBox Extension Pack.
3. Launch VirtualBox Manager and confirm it opens without error.

## Part B — Create the Host-Only Network

1. In VirtualBox Manager: **File → Tools → Network Manager** (labeled **Host Network Manager** in some versions).
2. Create a **Host-only Adapter** (e.g., `vboxnet0`) if one doesn't already exist.
3. Note its subnet — typically `192.168.56.0/24`. This network has no gateway and no route anywhere else.
4. Confirm with your instructor whether to use the adapter's built-in DHCP or the static addressing scheme shown above (`.101` for Kali, `.102` for Metasploitable).

## Part C — Import Kali Linux

1. **File → Import Appliance**, select the Kali image.
2. Allocate **at least 4 GB RAM / 2 vCPUs**.
3. Before booting, open **Settings → Network**:
  - **Adapter 1:** Enabled → Attached to **Host-only Adapter** → select the adapter from Part B.
  - **Adapters 2-4: Disabled.**
4. Double-check no adapter is set to **Bridged** or **NAT** — fix this now if it is.
5. Boot Kali, log in with the credentials your instructor provided, and open a terminal.
6. Confirm your address:

```
ip addr
```

You should see an address in the host-only range.

## Part D — Import Metasploitable 2

1. Import the appliance (or create a VM and attach the provided disk image).
2. Allocate **1 GB RAM**.
3. **Settings → Network → Adapter 1:** same Host-only Adapter as Kali. **Adapters 2-4: Disabled.**
4. Boot the VM and log in with the credentials your instructor provides.
5. Record its address:

```
ifconfig
```

6. **Do not patch or update this machine** — its unpatched state is the point of the exercise.

## Part E — Import OWASP BWA

1. Same import process: **1 GB RAM, Adapter 1 = Host-only, Adapters 2-4 disabled.**
2. Boot the VM and note the IP address shown on its console/login banner.
3. **Never attach this VM to any network other than the host-only adapter.**

## Part F — Verify Isolation and Connectivity

1. From Kali, ping both targets:

```
ping -c 4 192.168.56.102
ping -c 4 <OWASP_BWA_IP>
```

Both should reply.

2. From Kali, confirm you have **no route to the Internet**:

```
ping -c 2 8.8.8.8
```

This should **fail** — that failure is correct. If it succeeds, an adapter is still set to NAT or Bridged; go back and fix it.

3. Try loading any external website in a browser on Kali — it should also fail.

### Part G — Take Baseline Snapshots

1. Shut down each VM cleanly.
2. For **each** of the three VMs: right-click → **Snapshot → Take Snapshot**, name it `baseline-pre-attack`.
3. This is your rollback point for every future lab session.

## 6. Checkpoint — What to Submit

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1. `ip addr` output from Kali showing its host-only address
2. `ifconfig` output from Metasploitable showing its host-only address
3. The **failed** ping to `8.8.8.8` from Kali (your proof of isolation)
4. A screenshot of all three `baseline-pre-attack` snapshots in VirtualBox Manager

## 7. Troubleshooting

| Symptom                                     | Likely cause / fix                                                                                                                              |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| VM gets no address on the host-only network | Confirm Adapter 1 is attached to the <i>same</i> host-only adapter on all three VMs, and that DHCP is enabled or you've addressed it statically |
| Kali can't ping Metasploitable              | Confirm Metasploitable's Adapter 1 is Host-only (not NAT), and that the VM has fully finished booting                                           |
| Kali <i>can</i> reach the Internet          | An adapter is set to NAT or Bridged — recheck every adapter slot, not just Adapter 1                                                            |

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*Next session: Lab 2 — Verifying Your Toolset (Reconnaissance and a First Scan). Come with this environment running and your snapshots taken.*